

Lecture 14: We Should Have Higher Standards...

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Intro

- Last lecture: characteristics vs covariances, and whether time-varying betas can save factor models
- Today: a different critique
- What if the problem is not the factor, but the test assets?

- Last lecture: characteristics vs covariances, and whether time-varying betas can save factor models
- Today: a different critique
- What if the problem is not the factor, but the test assets?
- Lewellen, Nagel, and Shanken ([2010](#)): pricing the 25 portfolios is easy like Sunday morning
- This raises the bar for what should count as a successful asset-pricing test

- Part I: why the factor zoo exploded ([Fama and French 2008](#))
- Part II: why pricing the 25 portfolios is too easy ([Lewellen, Nagel, and Shanken 2010](#))
- Part III: how to raise the bar in empirical tests ([Lewellen, Nagel, and Shanken 2010](#))
- Part IV (next class): a modern reconciliation ([Kelly, Pruitt, and Su 2019](#))

Part I: We Should Have Higher Standards...

During the 2000s and early 2010s:

- Many strategies with large expected returns and no relation to betas appeared
- Examples: accruals, profitability, investment, asset growth, ...
- See Fama and French ([2008](#)) for an extensive analysis

The Factor-Zoo Explosion

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- Also: several papers proposing theories + factors that price the 25 portfolios

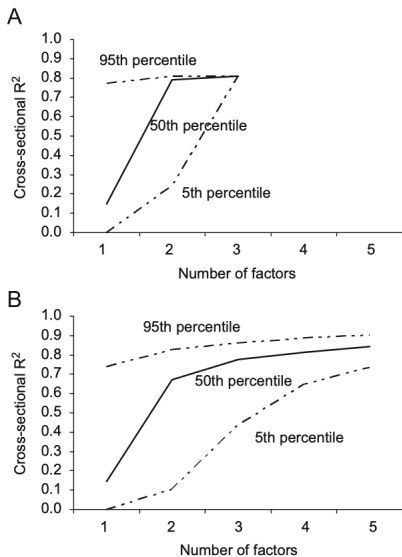
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- Lewellen, Nagel, and Shanken ([2010](#)): this is too easy!

- The 25 portfolios have a very strong factor structure
- We already know that MKT, SMB, and HML price them well
- If you give me a factor that is correlated with them... it will price the 25 portfolios too!
- We're judging models based on just a few assets!

A Killer Simulation



- Panel A: simulate random combinations of MKT, SMB, and HML
- Estimate the time-series betas and then run the cross-sectional pass
- Repeat many times and keep track of R^2
- Panel B: do the same, but with combinations of the 25 portfolios
- If you're not scared, look again

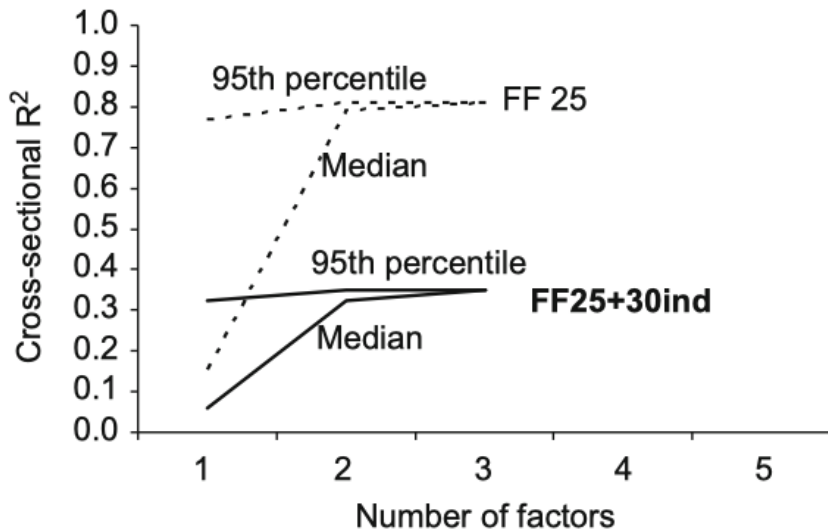
Questions?

Part II: How To Improve Tests

They provide a few ways to improve the tests:

1. Expand the set of test portfolios beyond size-B/M portfolios
2. Take the magnitude of the cross-sectional slopes seriously
3. Report the GLS R^2 instead of OLS R^2
4. If a factor is traded, it should price itself (i.e., have a significant risk premium)
5. Report confidence intervals for the cross-sectional R^2

One Example of Improvement



Questions?

- Fama, Eugene F., and Kenneth R. French. 2008. "Dissecting Anomalies." *The Journal of Finance* 63 (4): 1653–78. <https://doi.org/10.1111/j.1540-6261.2008.01371.x>.
- Kelly, Bryan, Seth Pruitt, and Yinan Su. 2019. "Characteristics Are Covariances: A Unified Model of Risk and Return." *Journal of Financial Economics* 134 (3): 501–24. <https://doi.org/10.1016/j.jfineco.2019.05.001>.
- Lewellen, Jonathan, Stefan Nagel, and Jay Shanken. 2010. "A Skeptical Appraisal of Asset Pricing Tests." *Journal of Financial Economics* 96 (2): 175–94. <https://doi.org/10.1016/j.jfineco.2009.09.001>.